MILITARY SPECIFICATION

VALVE, SOLENOID

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of 24-volt (V), solenoid actuated, hydraulic directional control valves with an operating pressure of 3000 pounds per square inch (psi) (see 6.1).

1.2 Classification. Valves shall be of the following types, as specified (see 6.2):

Type I	- Two position, four-way. Rated flow capacity: 6 gallons per minute (gpm).
Type II	- Two position, two-way. Rated flow capacity: 3.5 gpm.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed *Standardization Document Improvement Proposal* (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS
MILITARY

MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base for Preservation and Operation.

STANDARDS
FEDERAL

FED-STD-H28/2	- Screw Thread Standards for Federal Services Section 2 Unified Inch Screw Threads UN and UNR Thread Forms.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

9098233	- Valve, Solenoid (Type I).
9098673	- Valve, Solenoid (Type II).

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish valves which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the valves to be furnished to the Government. All subsequent valves delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free from defects which adversely affect performance or serviceability of the finished product. Exposed functional parts shall be fabricated from suitable corrosion resistant materials or treated to prevent corrosion (see 4.8.1 and 6.4).

3.2.1 Metals. All metals used in valve construction, except metal which is in constant contact with the hydraulic fluid, shall be of corrosion-resisting type or shall be adequately protected to resist corrosion during the normal service life of the valve and shall meet the requirements of 3.4.12.4 (see 4.8.1).

3.2.1.1 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion in accordance with requirements of MIL-STD-889 (see 4.8.1).

3.3 Design and construction. Unless otherwise specified (see 6.2), the valve shall be designed and constructed to the form and dimensions specified herein, and on the applicable drawing specified below (see 4.8.1 and 4.8.2):

Type I	- Drawing 9098233.
Type II	- Drawing 9098673.

3.3.1 Standard parts. Military standard parts shall be incorporated whenever possible. Commerical standard parts may be used, provided they are interchangeable with military standard parts without modification (see 4.8.1 and 4.8.2).

3.3.2 Interchangeability. The valve shall be mechanically and electrically interchangeable with all other such valves of the same military part number, previously furnished for this application by any manufacturer (see 4.8.1).

3.3.3 Threaded parts. Screw threads shall conform to FED-STD-H28/2. The form, number per inch, and class of threads shall be as specified on applicable standards or drawings (see 4.8.1 and 4.8.2).

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3.3.4 Locking devices. Lockwasher, self-locking nuts, or other approved locking devices shall be incorporated where specified, or where required, to prevent loosening of components (see 4.8.1 and 4.8.2).

3.3.5 Solenoid. The solenoid for operating the hydraulic valve shall be compact, designed for continuous or intermittent duty, and shall be provided with single coil windings. The solenoid shall be totally enclosed and shall be so designed that hydraulic fluid can at no time come in contact with the electrical windings (see 4.8.1 and 4.8.2).

3.3.5.1 Coils. The coils shall be evenly wound and insulated to meet the performance requirements specified herein. The coils shall be completely insulated from all other parts of the valve. Coils shall be suitably taped and impregnated, as required, to prevent damage under prolonged exposure to environmental conditions (see 4.8.1 and 4.8.2).

3.3.6 Weight. The gross weight of the valve shall not exceed 2.0 pounds (see 4.8.2).

3.4 Performance.

3.4.1 Operating time.

3.4.1.1 Type I. The valve shall operate (change port interconnection) in not more than 0.05 second, with 18 V direct current (dc) applied (see 4.8.3.1).

3.4.1.2 Type II. The valve shall operate (change port interconnection) in not more than 0.07 second, with 18 V dc applied (see 4.8.3.1).

3.4.2 Operating pressure. The valve shall operate with any pressure from 250 psi to 3000 psi applied to any port except return, and shall operate at any pressure within a pressure differential range, between the pressure port and return port, from 250 psi to 3000 psi (return port applicable to type I only) while using a hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083 (see 4.8.3.2).

3.4.3 Operating voltage. The valve shall operate at any voltage within a range of 18 to 30 V dc, with operating pressure applied (see 4.8.3.3).

3.4.4 Operating current. The valve shall require no more than 1.0 ampere (A) at 24 V dc to operate (see 4.8.3.3).

3.4.5 Rated flow and pressure drop.

3.4.5.1 Type I. The rated flow capacity of the valve using hydraulic fluid as specified in 3.4.2 shall be 6 gpm, through any interconnected ports, with a pressure drop across the valve to be not more than 150 psi (see 4.8.3.4).

3.4.5.2 Type II. The rated flow capacity of the valve using hydraulic fluid as specified in 3.4.2 shall be 3.5 gpm with a pressure drop across the valve to be not more than 60 psi (see 4.8.3.4).

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3.4.6 Proof pressure.

3.4.6.1 Type I. The valve shall withstand a proof pressure of 4500 psi, except the return port shall withstand a proof pressure of 1500 psi, without leakage or damage (see 4.8.3.5).

3.4.6.2 Type II. The valves shall withstand a proof pressure of 4500 psi without external leakage or damage (see 4.8.3.5).

3.4.7 Burst pressure.

3.4.7.1 Type I. The valve ports shall withstand hydraulic pressure of 7500 psi, except the return port shall withstand a pressure of 2500 psi without bursting (see 4.8.3.6).

3.4.7.2 Type II. The valve shall withstand hydraulic pressure of 7500 psi without bursting (see 4.8.3.6).

3.4.8 Leakage.

3.4.8.1 Pressure port.

3.4.8.1.1 Type I. The valve, in the de-energized position, shall withstand hydraulic pressures up to and including 3000 psi applied at the pressure port. Resultant combined internal leakage from cylinder number 2 port and return port shall be not greater than 10 cubic centimeters per minute (cc/min), with no external leakage (see 4.8.3.7.1.1).

3.4.8.1.2 Type II. The valve, in the energized position, shall withstand hydraulic pressures up to and including 3000 psi, applied at one port, with resultant internal leakage from the other port not in excess of 5 cc/min (see 4.8.3.7.1.2).

3.4.8.2 Return port (type I only). The valve, in the de-energized position, shall withstand application of hydraulic pressure of 10 psi at the return port and cylinder number 2 port without external leakage (see 4.8.3.7.2).

3.4.9 Endurance. The valve shall withstand 20 000 cycles of operation at 3000 psi hydraulic pressure, without damage. A cycle of operation shall consist of operation of the valve from the de-energized position to the energized position, and back to the de-energized position. The valve shall subsequently meet the requirements of 3.4.2 through 3.4.6, and 3.4.8 (see 4.8.3.8).

3.4.10 Solenoid temperature rise. The valve solenoid shall not exceed a temperature rise greater than 39 degrees Fahrenheit (°F) per minute of continuous operation, when used with voltages specified herein (see 4.8.3.9).

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3.4.11 Dieletric strength. The solenoid shall withstand application of 1000 V root mean square (rms), 60 hertz (Hz) potential, between terminals and ground as specified in MIL-STD-202, method 301, without flashover. Insulation shall evidence no loosening, cracking, charring, burning, smoking, rupture, or other damage resulting in a reduction in dielectric strength (see 4.8.3.10).

3.4.12 Environmental.

3.4.12.1 High temperature. The valve shall withstand exposure to a high temperature of 160 ± 5°F in accordance with method 501.2 procedure II of MIL-STD-810, without damage, and shall conform to performance requirements of to 3.4.2, 3.4.3, 3.4.5, and 3.4.8 during exposure. Subsequent performance shall conform to requirements of 3.4.9 (see 4.8.3.11.1).

3.4.12.2 Low temperature. The valve shall withstand exposure to a low temperature of -65 (+5, -0)°F in accordance with method 502.2, procedure II of MIL-STD-810, without damage, and shall conform to requirements of 3.4.2, 3.4.3, 3.4.5, and 3.4.8 during exposure. Subsequent performance shall conform to the requirements of 3.4.9 (see 4.8.3.11.2).

3.4.12.3 Fungus. The valve shall withstand exposure to fungus in accordance with method 508.3 of MIL-STD-810. Subsequent performance shall conform to 3.4.2, 3.4.3, 3.4.5, and 3.4.8 (see 4.8.3.11.3).

3.4.12.4 Salt spray. The valve shall withstand exposure to salt spray in accordance with MIL-STD-202, method 101, except duration of test shall be 200 hours, with subsequent performance conforming to 3.4.3, 3.4.3, 3.4.5, and 3.4.8 (see 4.8.3.11.4).

3.4.12.5 Sand and dust. The valve shall withstand exposure to sand and dust in accordance with method 510.2 of MIL-STD-810, except the temperature shall be not more than 125°F for 24 hours, without damage. Subsequent performance shall conform to 3.4.2, 3.4.3, 3.4.5, and 3.4.9 (see 4.8.3.11.5).

3.4.12.6 Shock. The valve shall withstand exposure to shock in accordance with method 516.3, procedure I of MIL-STD-810, without damage or loosening of components. Subsequent performance shall conform to 3.4.2, 3.4.3, 3.4.5, and 3.4.9 (see 4.8.3.11.6).

3.4.12.7 Vibration. The valve shall withstand vibration in accordance with MIL-STD-202, except vibration shall be for 1 hour at 77°F, and 30 minutes at minus 65°F. Vibration shall be applied in the direction of each of the three major axes, without damage or loosening of components. Subsequent performance shall conform to 3.4.2, 3.4.3, 3.4.5, and 3.4.9 (see 4.8.3.11.7).

3.4.12.8 Humidity. The valve shall withstand exposure to humidity in accordance with method 507.2 of MIL-STD-810, and with subsequent performance conforming to 3.4.5 and 3.4.6 (see 4.8.3.11.8).

3.4.12.9 Electromagnetic compatibility. The valve solenoid shall conform to electromagnetic compatibility requirements of class C1, group 1 of MIL-STD-461 for tactical equipment (see 4.8.3.11.9).

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3.5 Marking. The valve shall be marked in accordance with MIL-STD-130 and as specified on the applicable military drawing or standard. When specified on the applicable drawing or standard, nameplate shall be in accordance with MIL-P-514. Plate to be made of photographic aluminum (sensitized), not less than 0.030 inch thick, with a black background. Marking shall include the following information, in the order shown (see 4.8.2):

a. Valve, Solenoid
b. Military part number
c. Manufacturer's serial number
d. Manufacturer's name
e. Date of manufacture
f. Contract or order number
g. US

3.6 Finish. Unless otherwise specified (see 6.2), finish shall conform to requirements specified on the applicable drawing (see 4.8.2).

3.7 Workmanship. Workmanship shall be of a quality which will assure a product free of burrs, rust, scratches, chips, sharp edges, loose or defective connectors, cracked insulation, faulty soldering, or other defects which will affect serviceability or appearance (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply wih all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 77 $\pm$ 15°F.
b. Barometric pressure 28.5 + 2.0 inches mercury (Hg).
c. Relative humidity 50 $\pm$ 30 percent.

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply valves that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of four valves. Preproduction inspection shall consist of inspection as specified in table I.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select four valves, from the first 10 valves produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table I.

4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are

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similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE I. Sequence of first article tests.

Title	Requirement	Inspection	Sample No.
Materials and construction	3.2 thru 3.3.5.1	4.8.1	1 thru 4
Defects (see	3.3, 3.3.1, 3.3.3 thru 3.3.6, 3.5 thru 3.7	4.8.2	
Operating time	3.4.1	4.8.3.1	1
Operating pressure	3.4.2	4.8.3.2	
Operating voltage	3.4.3	4.8.3.3	
Operating current	3.4.4	4.8.3.3	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Solenoid temperature rise	3.4.10	4.8.3.9	
Electromagnetic compatibility	3.4.12.9	4.8.3.11.9	
Dielectric strength (60 second)	3.4.11	4.8.3.10	
Endurance	3.4.9	4.8.3.8	2
Operating pressure	3.4.2	4.8.3.2	
Operating voltage	3.4.3	4.8.3.3	
Operating current	3.4.4	4.8.3.3	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	
Leakage	3.4.8	4.8.3.7	
High temperature	3.4.12.1	4.8.3.11.1	3
Operating pressure	3.4.2	4.8.3.2	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	
Low temperature	3.4.12.2	4.8.3.11.2	
Operating pressure	3.4.2	4.8.3.2	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	
Sand and dust	3.4.12.5	4.8.3.11.5	
Operating pressure	3.4.2	4.8.3.2	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	
Fungus	3.4.12.3	4.8.3.11.3	
Operating pressure	3.4.2	4.8.3.2	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	

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TABLE I. Sequence of first article tests - Continued.

Title	Requirement	Inspection	Sample No.
Shock	3.4.12.6	4.8.3.11.6	4
Operating pressure	3.4.2	4.8.3.2	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	
Vibration	3.4.12.7	4.8.3.11.7	
Operating pressure	3.4.2	4.8.3.2	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	
Humidity	3.4.12.8	4.8.3.11.8	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	
Salt spray	3.4.12.4	4.8.3.11.4	
Operating pressure	3.4.2	4.8.3.2	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the valves of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 20 consecutively produced assemblies that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs.

4.5.1.3 Sampling for test. Samples for tests shall be selected in accordance with level S-3 of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AOL).

Classification	AOL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

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TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability not within tolerance (see 3.3).	SIE 1/
102	Threads not as specified (see 3.3.3).	SIE
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability not within tolerance (see 3.3).	SIE
202	Standard parts not as specified (see 3.3.1).	Visual and SIE
203	Loose fasteners (see 3.3.4).	Visual and tactile
204	Solenoid not as specified (see 3.3.5).	Visual
205	Coil not as specified (see 3.3.5.1).	Visual
206	Weight greater than specified (see 3.3.6).	SIE
207	Incorrect or illegible marking (see 3.5).	Visual
208	Finish not as specified (see 3.6).	Visual
209	Faulty workmanship (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Test. Each valve selected in accordance with 4.5.1.3 shall be subjected to the quality conformance tests specified in table III in the numerical order listed, using and AQL of 6.5 on the basis of percent defective.

TABLE III. Quality conformance tests.

Title	Requirement	Test
Dielectric strength (1 second)	3.4.11	4.8.3.10
Solenoid temperature rise	3.4.10	4.8.3.9
Proof pressure	3.4.6	4.8.3.5
Rated flow and pressure drop	3.4.5	4.8.3.4
Leakage	3.4.8	4.8.3.7
Operating pressure	3.4.2	4.8.3.2

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4.6 Control tests. Unless otherwise specified (see 6.2), two valves from each 500 produced shall be selected for control tests specified in table IV. The tests shall be performed in the order shown.

TABLE IV. Control tests.

Title	Requirement	Test	Sample No.
Defects	3.3, 3.3.1, 3.3.3 thru 3.3.6, 3.5 thru 3.6	4.8.2	
Operating time	3.4.1	4.8.3.1	1
Burst pressure	3.4.7	4.8.3.6	
Defects	3.3, 3.3.1, 3.3.3 thru 3.3.6, 3.5 thru 3.6	4.8.2	
Operating voltage	3.4.3	4.8.3.3	2 1/
Operating current	3.4.4	4.8.3.3	
Environmental	3.4.12	4.8.3.11	
Operating pressure	3.4.2	4.8.3.2	
Rated flow and pressure drop	3.4.5	4.8.3.4	
Proof pressure	3.4.6	4.8.3.5	

1/ Each even numbered sample shall receive one environmental test, sequentially in the order listed in 4.8.3.11.

4.7 Failure. Failure of any valve to pass any of the quality conformance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3.5.1 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.3.1, 3.3.3 through 3.3.6, 3.5 through 3.7, shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Performance. Unless otherwise specified (see 6.2), performance testing shall be conducted under the following conditions:

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a. Fluids. All tests shall be performed with hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083. The hydraulic fluid temperature shall be 90 ± 20°F.
b. Cleaning. All grease or foreign material shall be removed from interior parts of the valve before testing.
c. Air removal. Air shall be completely removed from all hydraulic portions of the valves prior to tests.

4.8.3.1 Operating time. To determine conformance to 3.4.1, the valve shall be mounted and electrically connected as in intended use, with 18 V dc applied. The valve shall be operated (change port interconnection), and the time required recorded.

4.8.3.1 Operating pressure. To determine conformance to 3.4.2, the valve shall pass the tests of 4.8.3.2.1 through 4.8.3.2.3, as applicable.

4.8.3.2.1 Pressure port.

4.8.3.2.1.1 Type I. The valve shall be subjected to hydraulic pressure at the pressure port, with cylinder number 1 and 2 ports connected independently to loading valve and flow meter, and the return port open. With 250 psi maintained at the pressure port, the valve shall actuate with 18 V dc or less applied to the solenoid. The pressure drop through the valve, either energized or de-energized, shall be not more than 150 psi at 6.0 gpm flow. With 3000 psi maintained at the pressure port, the valve shall actuate with application of 18 V dc or less, and the pressure drop through the valve, either energized or de-energized, shall be not more than 150 psi at 6.0 gpm (see 3.4.2, 3.4.3 and 3.4.5.1).

4.8.3.2.1.2 Type II. The valve in the energized position shall be subjected to a hydraulic pressure of 3000 psi at port "A", and port "B" connected to a flow meter (see 3.4.2). The valve shall then be energized, with an application of 18 V dc, or less, actuating the solenoid (see 3.4.3). With the valve de-energized and a flow through the valve of 3.5 gpm, the pressure drop across the valve shall be recorded to determine conformance to 3.4.5. Test shall be repeated with set-up at ports "A" and "B" reversed.

4.8.3.2.2 Return port (type I only). The valve shall be energized with 18 V dc, or less, at a pressure differential of 250 psi between the pressure port and the return port (pressure port at the greater pressure). A pressure of 1000 psi and a flow of 6 gpm shall then be applied at the return port. The pressure drop between the return port and cylinder number 1 port shall be not more than 150 psi, with a flow of 6 gpm in either direction. The test shall be repeated, except that 1000 psi shall be applied at cylinder number 1 port (see 3.4.2, 3.4.3 and 3.4.5).

4.8.3.2.3 Actuation. With a hydraulic pressure of 3000 psi at the pressure port and the valve operating as in 4.8.3.2.1, the valve shall be actuated with 24 V dc no less than 10 times to assure that there is no indication of binding, chatter, or other unsatisfactory operation.

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4.8.3.3 Operating voltage and current. During testing, to determine conformance to 3.4.3, voltage shall be varied between 18 and 30 V dc and operation observed. To determine conformance to 3.4.4, current draw shall be measured during operation at 24 V dc.

4.8.3.4 Rated flow and pressure drop. To determine conformance to 3.4.5, the rated flow shall be accurately maintained at the value specified in 3.4.5. A manometer or other accurate means of measuring pressure drop through interconnected ports shall be employed. The pressure drop shall be measured and recorded.

4.8.3.5 Proof pressure. To determine conformance to 3.4.6, the valve shall be subjected to the pressure specified in 3.4.6 at the pressure port and the return port (type I only) for a period of 2 minutes at each application. The test shall be performed with the solenoid in both the energized and de-energized position. For type I valves, when testing the pressure port, the return port shall be open. The valve shall be electrically actuated after application of pressure. There shall be no evidence of external leakage, failure, or permanent deformation.

4.8.3.6 Burst pressure. After completion of all other tests, the valve shall be subjected to the pressure specified in 3.4.7 at the pressure and return port (type I only) for a period of 3 minutes at each application. The test shall be performed with the solenoid in both the energized and de-energized positions. For type I valves, when testing the pressure port, the return port shall be open. The rate of application of pressure shall not be more than 26 000 psi per minute. The valve need not be operable after testing. Results shall be recorded to determine conformance to 3.4.7.

4.8.3.7 Leakage.

4.8.3.7.1 Pressure port.

4.8.3.7.1.1 Type I. To determine conformance to 3.4.8.1.1, with the valve de-energized, a hydraulic pressure of 5 to 10 psi shall be applied at the pressure port, and maintained for a period of 2 minutes. Leakage measurements (external and internal) shall be taken and shall not be more than the values specified in 3.4.8.1.1. Pressure shall then be raised to 3000 psi, and maintained for a period of 2 minutes. Leakage measurements shall be recorded.

4.8.3.7.1.2 Type II. To determine conformance to 3.4.8.1.2, with the valve de-energized, the valve shall be subjected to a hydraulic pressure of 3000 psi at port "A" and energized with an application of the not more than 18 V dc, actuating the solenoid. A hydraulic pressure of 3000 psi at port "A" shall be maintained for a period of 2 minutes, while the leakage at port "B" is recorded. The hydraulic pressure at port "A" shall then be reduced to 5 to 10 psi and maintained for a period of 2 minutes, while the leakage at port "B" is recorded. The pressure set-up at ports "A" and "B" shall be reversed and test repeated.

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4.8.3.7.2 Return port (type I only). With the valve de-energized and with the pressure port and cylinder number 1 port plugged, a hydraulic pressure of 5 to 10 psi shall be applied at the return port and cylinder number 2 port. After test, the valve shall be examined to determine conformance to 3.4.8.2.

4.8.3.8 Endurance. To determine conformance to 3.4.9, the valve shall be solenoid operated for 20 000 cycles, 50 percent of which shall be with the solenoid voltage at 30 V dc, with a hydraulic pressure of not less than 3000 psi at the pressure port. A restriction shall be placed in the line from each cylinder port to maintain not less than a 3000 psi at the cylinder port interconnected to the pressure port. For type I valve, the pressure at the opposite cylinder port shall drop to zero psi before proceeding to the next cycle. After test, the valve shall pass the tests specified in 4.8.3.2, 4.8.3.4, 4.8.3.5 and 4.8.3.7.

4.8.3.9 Solenoid temperature rise. To determine conformance to 3.4.10, the valve solenoid shall be subjected to 30 V dc for a continuous 3 minutes, at an ambient temperature of 77 $\pm$ 15°F.

4.8.3.10 Dielectric strength. To determine conformance to 3.4.11, the solenoid shall be subjected to a test voltage of 1000 V rms at a frequency of 60 Hz between terminals and ground. Application shall be as specified in MIL-STD-202, method 301, for 1 second for production units, and for 60 seconds for preproduction test units. Flashover during this test, or a noticeable decrease in insulation resistance as a result of this test, shall constitute failure.

4.8.3.11 Environmental.

4.8.3.11.1 High temperature. To determine conformance to 3.4.12.1, the valve shall be subjected to the high temperature test specified in procedure II of MIL-STD-810, method 501.2. High temperature testing shall be conducted at the temperature specified in 3.4.12.1. While at the high temperature, the valve shall be subjected to tests specified in 4.8.3.2, 4.8.3.4, and 4.8.3.5. After the high temperature testing the valve shall be stabilized at room temperature, examined, and tested as specified in 4.8.3.8 (50 cycles only). Test results shall be recorded.

4.8.3.11.2 Low temperature. To determine conformance to 3.4.12.2, the valve shall be subjected to the low temperature tests specified in procedure II of MIL-STD-810, method 502.2, except the valve shall be maintained at a low temperature of not nore than minus 65°F. While at the low temperature the valve shall be tested as specified in 4.8.3.2, 4.8.3.4, and applicable tests of 4.8.3.5. After low temperature exposure, the valve shall be stabilized at room temperature, examined, and tested as specified in 4.8.3.8 (50 cycles only). Test results shall be recorded.

4.8.3.11.3 Fungus. To determine conformance to 3.4.12.3, the valve shall be subjected to the fungus resistance test of MIL-STD-810, method 508.3. After the test, the valve shall pass the tests as specified in 4.8.3.2, 4.8.3.4, and 4.8.3.5.

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4.8.3.11.4 Salt spray. To determine conformance to 3.4.12.4, the valve shall be subjected to the salt spray test of MIL-STD-202, method 101, except the test duration shall be 200 hours. After the test, the valve shall pass the tests as specified in 4.8.3.2, 4.8.3.4, and 4.8.3.5.

4.8.3.11.5 Sand and dust. To determine conformance to 3.4.12.5, the valve shall be subjected to sand and dust resistance test of MIL-STD-810, method 510.2, except the test shall be conducted at an ambient temperature of 120 ± 5°F only, and the test period shall be 24 hours. The valve shall be mounted in the dust chamber as in intended use. After the test, the valve shall pass the tests specified in 4.8.3.2, 4.8.3.4, and 4.8.3.5.

4.8.3.11.6 Shock. To determine conformance to 3.4.12.6, the valve shall be subjected to the shock test specified in MIL-STD-810, method 516.3. After the test, the valve shall be examined and shall pass the tests specified in 4.8.3.2, 4.8.3.4, and 4.8.3.5.

4.8.3.11.7 Vibration. To determine conformance to 3.4.12.7, the valve shall be subjected to the vibration test specified in MIL-STD-202, method 201, except vibration sahll be applied for 1 hour at 77 ± 15°F and 30 minutes at minus 65 ± 5°F, in the direction of each of the three major axes. Following the vibration test, the valve shall be examined and shall pass the tests specified in 4.8.3.2, 4.8.3.4, and 4.8.3.5.

4.8.3.11.8 Humidity. To determine conformance to 3.4.12.8, the valve shall be subjected to the humidity test specified in MIL-STD-810, method 507.2, procedure III. After the test, the valve shall pass the tests specified in 4.8.3.4 and 4.8.3.5.

4.8.3.11.9 Electromagnetic compatibility. To determine conformance to 3.4.12.9, the valve shall be tested as specified in MIL-STD-462.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. These valves are intended for use with loader-transporter, guided missile (Hawk Loader).

6.1.1 Type I. The valve covered by this specification is a two position four-way valve, used to automatically select the fluid path in a 3000 psi hydraulic system to reduce the final cylinder actuating pressure.

6.1.2 Type II. The valve covered by this specification is a two position, two-way valve used to interrupt the fluid flow in a 3000 psi hydraulic system, or permit flow in either direction through the valve.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of valve required (see 1.2).
c. If first article samples are not required (see 3.1).
d. If construction shall be other than as specified (see 3.3).
e. If finish shall be other than as specified (see 3.6).
f. If responsibility for inspection shall be other than as specified (see 4.1).
g. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
h. If inspection temperature conditions shall be other than as specified (see 4.3).
i. If preproduction inspection is required (see 4.4.1).
j. If initial production inspection is not required (see 4.4.2).
k. If the sampling plan for control testing shall be other than as specified (see 4.6).
l. If control tests are not required (see 4.6).
m. If testing conditions shall be other than as specified (see 4.8.3).
n. Applicable level of preservation, packaging, packing and marking (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.4 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
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